

Lab 2: Time series decomposition

Trend and seasonality on a synthetic series and on Atlanta temperatures

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2025–2026

Goals

- Implement and compare three trend estimators (centred moving average, polynomial regression, LOESS).
- Implement and compare three seasonality estimators (seasonal averaging, dummy regression, Fourier model).
- Apply the full workflow to the Atlanta monthly temperature dataset.
- Compare additive and multiplicative decomposition models.

Setup

- Python 3 with `numpy`, `pandas`, `scipy`, `statsmodels`, `matplotlib`.
- Reproducibility: `numpy.random.seed(42)`.
- Dataset: `data/AvTempAtlanta.txt`.
- Skeleton: `seed.py` provides function signatures.

Exercise 1 – Synthetic decomposition

Generate $T = 240$ months (20 years) of a synthetic series:

$$Y_t = \underbrace{20 + 0.05t}_{\text{trend}} + \underbrace{5 \sin(2\pi t/12)}_{\text{seasonality}} + \underbrace{\varepsilon_t}_{\mathcal{N}(0,1)}.$$

- (1) Plot Y_t along with the true components on a 4-panel figure (observed Y_t , true T_t , true S_t , true ε_t).
- (2) Comment on the visual signature of the additive model (constant seasonal amplitude).

Exercise 2 – Trend estimation: three methods

- (1) Implement the centred moving average `centred_ma(y, k)` of window $2k + 1$.
- (2) Apply three methods to Y_t from Exercise 1:
 - centred MA with $k = 6$ (window 13);
 - linear regression (`sklearn.LinearRegression`);
 - LOESS (`statsmodels.nonparametric.smoothers_lowess`, `frac=0.2`).

Plot the three estimated trends together with the true trend $\hat{T}_t = 20 + 0.05t$.

(3) Compute the RMSE of each estimator versus the true trend. Comment.

Exercise 3 – Seasonality estimation: three methods

(1) Detrend Y_t from Exercise 1 using LOESS, then estimate the seasonality by:

- seasonal averaging (12 monthly indices);
- OLS regression with 11 dummy variables;
- cosine-sine model with one harmonic ($\omega = 2\pi/12$).

(2) Plot the three estimated seasonal patterns ($\hat{S}_k, k = 1, \dots, 12$) along with the truth $5 \sin(2\pi k/12)$.

(3) Compute the RMSE of each estimator versus the truth. Which method performs best, and why?

Exercise 4 – Application to Atlanta temperatures

Load `data/AvTempAtlanta.txt` (138 years $\times$ 12 months = 1656 observations, time index $t = 1, \dots, 1656$).

(1) Plot the series. What is the seasonal period? Does the amplitude look constant over time?

(2) Decompose using the workflow:

- estimate the trend with a 2×12 centred MA;
- compute the detrended series;
- estimate seasonality by averaging (12 monthly indices, re-centred);
- compute the residual.

Plot the four components on a 4-panel figure.

(3) Compare with `statsmodels.tsa.seasonal_decompose(y, period=12, model="additive")`. Do you get the same components?

Exercise 5 – Bonus: additive vs multiplicative

(1) Generate a multiplicative series:

$$Y_t = (20 + 0.1t) \cdot (1 + 0.3 \sin(2\pi t/12)) \cdot e^{\eta_t}, \quad \eta_t \sim \mathcal{N}(0, 0.05).$$

Plot it. Comment on the seasonal amplitude.

(2) Apply the additive workflow on Y_t vs. the additive workflow on $\log Y_t$. Compare the residuals.

(3) Conclude: when should you use the multiplicative model?

Warning

For each plot, save a PNG (`lab2_ex1_synth.png`, etc.). Set the seed at the top of every script.